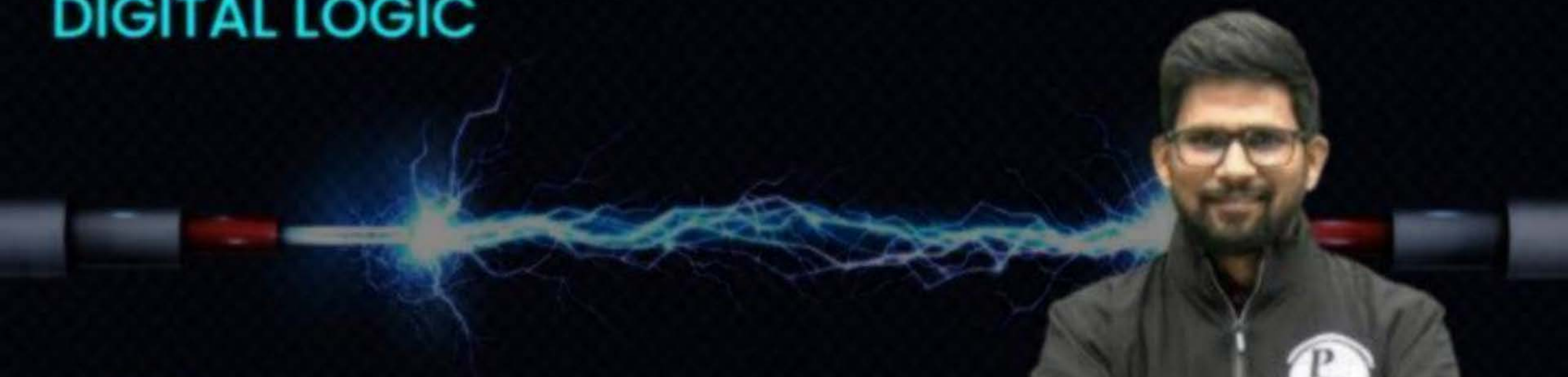
COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No. 09

BOOLEAN THEOREMS AND GATES



By- Chandan Gupta Sir



NAND gate



- NAND gate cont.

H·W

Minimum no. of 2 i/P NAND gate required to implement

1. $f(A,B,C) = A\bar{B} + B\bar{C} + A\bar{C}$ → 4 NAND → $A\bar{B} + B\bar{C} = (\bar{B} + \bar{C})(B + A)$

2. $f(A,B,C) = \bar{A}\bar{B} + \bar{B}\bar{C}$ → redundant 4 NAND

$= \overline{B \cdot C}\,(B + A)$

$= R(B + A)$ → 1NAND

3. $f(A,B,C) = AB + BC + AC$

4. $f(A,B,C) = \overline{AB\bar{C} \cdot D}$

$f = RB + RA$

$\bar{f} = \overline{RB} \cdot \overline{RA}$

$\overline{RB}$ → P (1NAND)

$\overline{RA}$ → Q (1NAND)

$\bar{f} = P \cdot Q$

$f = \overline{P \cdot Q}$ → 1NAND

$\overline{R}$ $(B+A)$

$\overline{R}$ → 1NAND

$(B+A)$ → W ⇒ 3NAND

$f = R \cdot W$ → 2NAND

$f = \overline{A}\,\overline{B} + \overline{B}\,\overline{C}$

$f = \overline{B}[\overline{A}+\overline{C}] = \overline{B} \cdot \overline{A \cdot C}$

$\overline{A \cdot C}$ → Z → 1NAND

$f = \overline{B} \cdot Z$

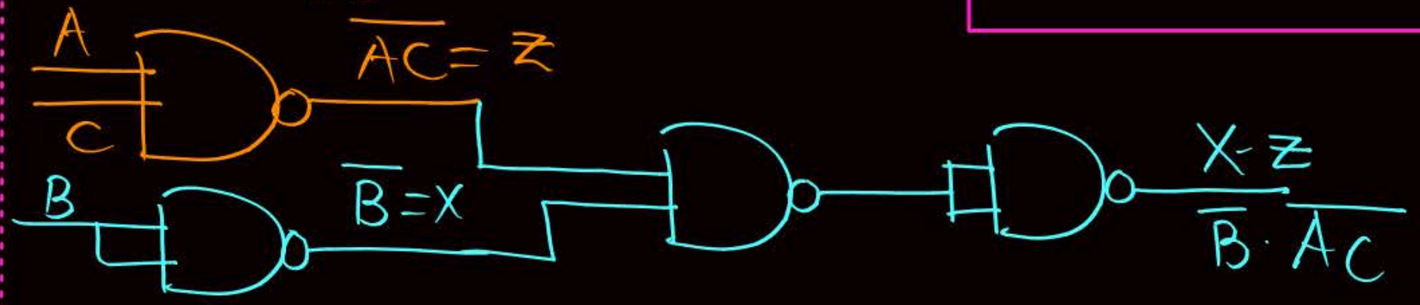
$\overline{B}$ → 1NAND → X

$f = X \cdot Z$ → 2NAND

$A+B$ $(+)$ → 3NAND

$A \cdot B$ AND $(\cdot)$ → 2NAND

$\overline{A \cdot B}$ → 1NAND



- $f(A,B,C) = AB + BC + CA$

$$\overline{f(A,B,C)} = \overline{AB+BC+CA} = \overline{AB} \cdot \overline{BC} \cdot \overline{CA}$$

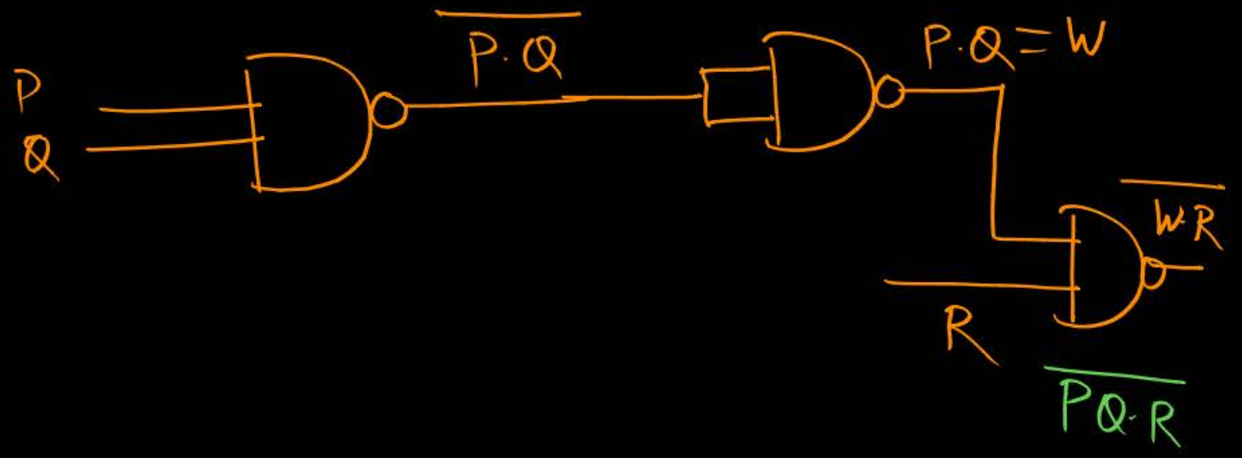
$\overline{AB}$ → P (1 NAND) $\quad \overline{BC}$ → Q (1 NAND) $\quad \overline{CA}$ → R (1 NAND)

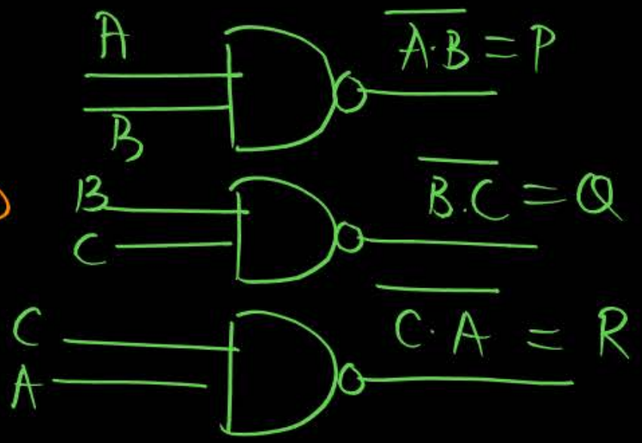
$$\overline{f} = P \cdot Q \cdot R$$

$$f = \overline{P \cdot Q \cdot R} = \overline{W \cdot R}$$

$\overline{W \cdot R}$ → 1 NAND

$P \cdot Q = W$ → 2 NAND





- $f(A,B,C,D) = \overline{(\overline{ABC})D} = \overline{W \cdot D}$ → 6 NAND

  → 1 NAND

$ABC = W$ → 5 NAND

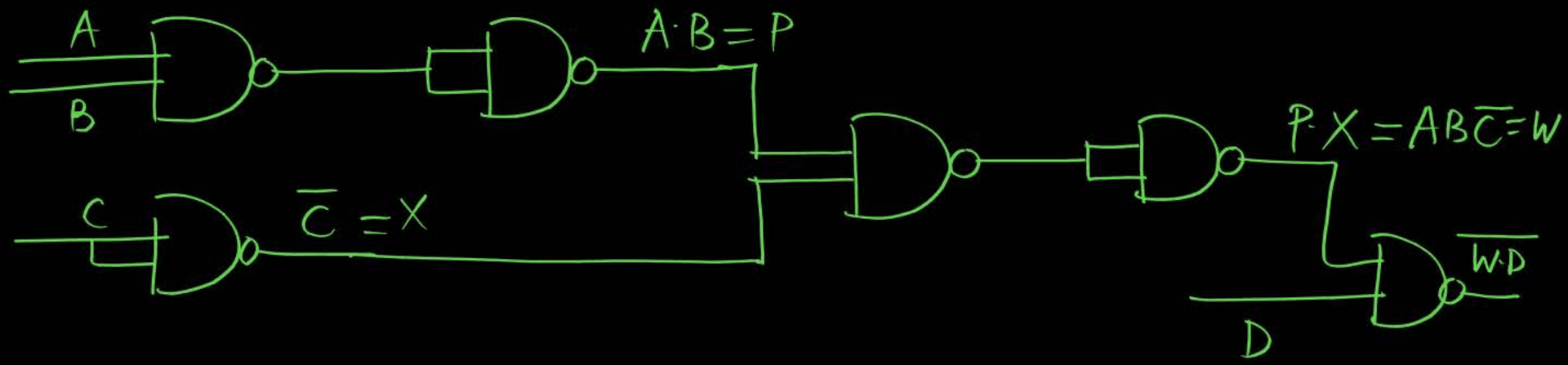
$A \cdot B \rightarrow P$

2 NAND

$P \cdot \overline{C} = W$

1 NAND)

2 NAND



- $f(A,B,C) = \overline{AB} + \overline{BC} = \overline{A}+\overline{B}+\overline{B}+\overline{C} = \overline{A}+\overline{B}+\overline{C} = \overline{ABC}$

Minimum no. of 2i/p NAND gate required ——— .

$\overline{f} = \overline{\overline{AB} + \overline{BC}} = AB \cdot BC = ABC$

$f = \overline{ABC}$

- $f(A,B,C,D) = \overline{AB} + \overline{CD}$ 5

$\overline{f} = AB \cdot CD = ABCD$

$f = \overline{ABCD} = \overline{W \cdot D}$ → 1 NAND

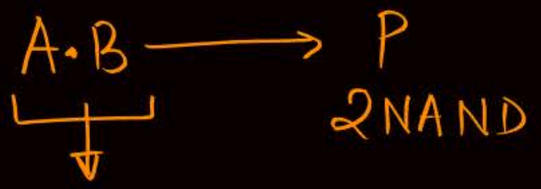
$W = A \cdot BC = P \cdot C$ → 2 NAND

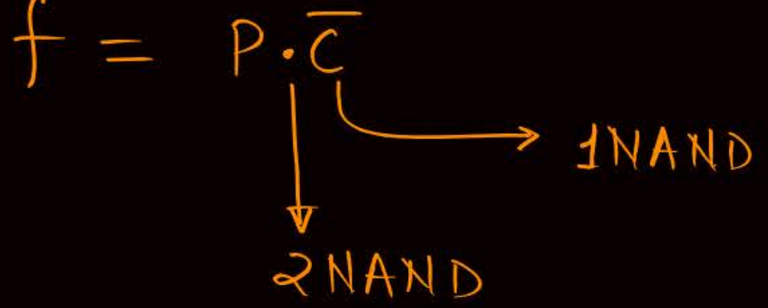
P → 2 NAND

$f = \overline{ABCD} = \overline{P \cdot Q}$ → 1 NAND

P → 2 NAND, Q → 2 NAND

$f(A,B,C) = A.B\bar{C}$ ⟶ minimum no. of 2-i/p NAND gate required to implement f 5.





Q. $f(A,B,C) = \bar{A}\bar{B} + A\bar{B} + BC$

Minimum no. of 2-i/p NAND gate required to implement $f(A,B,C)$ ____6____.

$$\bar{f} = \left(\overline{\bar{A}\cdot\bar{B}} \quad \overline{A\bar{B}}\right) \quad \overline{BC}$$

$\overline{\bar{A}\cdot\bar{B}} = (A+B)(\bar{A}+\bar{B})$ → P (3 NAND)

$\overline{A\bar{B}}$ → 1 NAND → Q

$\overline{BC}$ → 1 NAND → R

$$f = \overline{P\cdot Q\cdot R} \longrightarrow 3 \text{ NAND}$$

$$f(A,B,C) = A \odot B + BC$$

$$\bar{f} = \overline{A \odot B + BC} = \overline{A \odot B} \cdot \overline{BC}$$

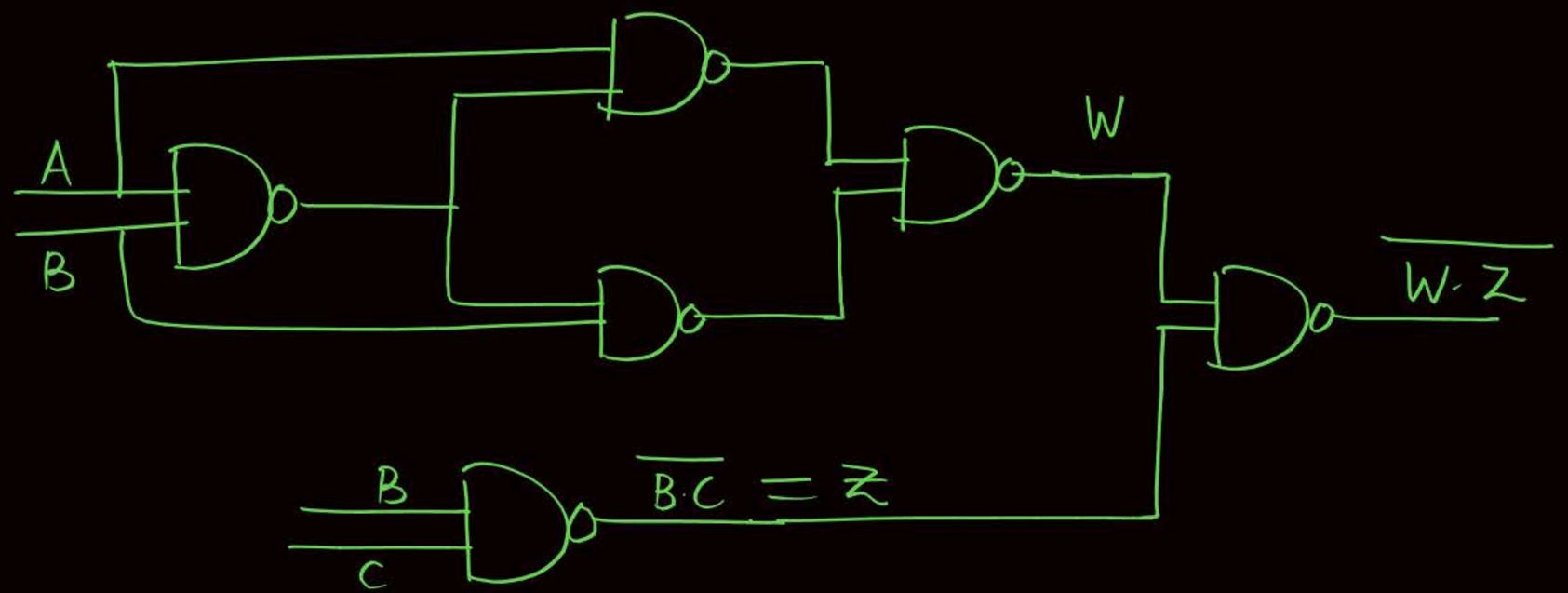
$$\bar{f} = (A \oplus B) \cdot \overline{BC}$$

$(A \oplus B)$ → 4 NAND → W

$\overline{BC}$ → 1 NAND (Z)

$$\bar{f} = W \cdot Z$$

$$f = \overline{W \cdot Z} \longrightarrow 1 \text{ NAND}$$


A
B
W
$\overline{W \cdot Z}$
B
C
$\overline{B \cdot C} = Z$

Question

$y = A\bar{B} + BC$

(H.W).

Minimum no. of 2-input NAND gates required to implement y _____

Question

$f(A, B, C) = \bar{A} + \bar{A}B + \bar{A}B\bar{C} = \bar{A}[1 + B + B\bar{C}] = \bar{A} \cdot 1 = \bar{A}$

Minimum no. of 2-input NAND gates required to implement y 1

Question

$y = \bar{A}BC$

To implement y, minimum no. of 2-input NAND gates required is _____

H.W.

Question

$f(A, B, C, D) = (\bar{A} + \bar{B})(C + D)$

Minimum no. of 2-input NAND gates required to implement $f(A, B, C, D)$ 4

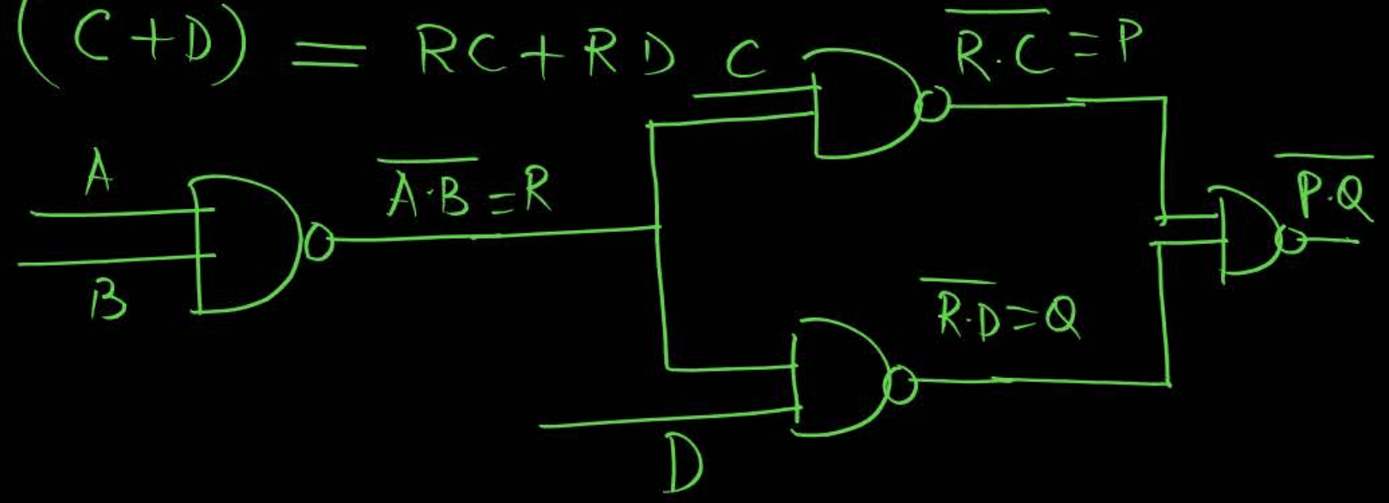
$= \overline{AB}\,(C+D) = R(C+D) = RC + RD$ → 1 NAND

$\bar{f} = \overline{RC} \cdot \overline{RD}$ → 1 NAND Q

$\overline{RC}$ → 1 NAND P

$\bar{f} = P \cdot Q$

$f = \overline{P \cdot Q}$ → 1 NAND



#Q. We have to implement two functions using minimum no of 2-i/p NAND gates:

(i). $f_1(A,B) = \bar{A}B + A\bar{B} = PB + AQ \longrightarrow \bar{f} = \overline{PB} \quad \overline{AQ}$

$W(1) \quad Z(1)$

$f = \overline{W.Z}$ 1 NAND

(ii) $f_2(A,B) = \bar{A}\bar{B} + A.B = PQ + AB$

Compliments of variables A & B are available, then minimum of 2-i/p NAND gate required in case (i) 3 & in case (ii) 3.

A, B, $\bar{A}$ (P), $\bar{B}$ (Q)

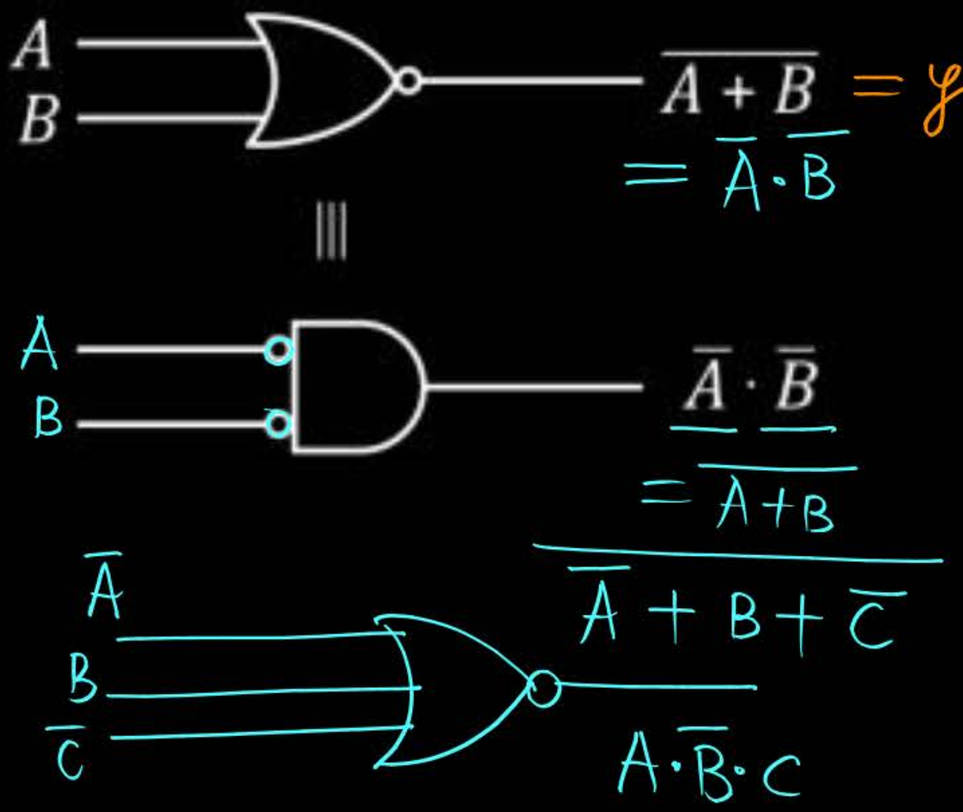
$\bar{f} = \overline{\bar{A}.B} \quad \overline{A.\bar{B}}$

NOR GATE

OR + NOT



Representation :



	2^1	2^0	
	A	B	$y = \overline{A+B}$
0	0	0	1
1	0	1	0
2	1	0	0
3	1	1	0

$y(A, B) = \Sigma(0)$

$\overline{A} \cdot \overline{B} = \pi(1,2,3)$

$= (A+\overline{B})(\overline{A}+B)$

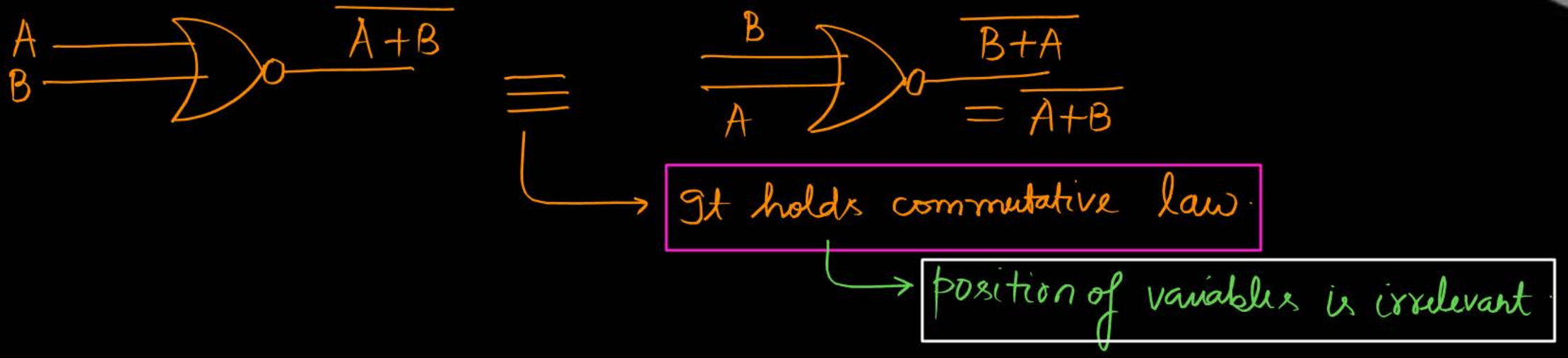
$(\overline{A}+\overline{B})$

$= (\overline{A} + B\cdot\overline{B})(A+\overline{B})$

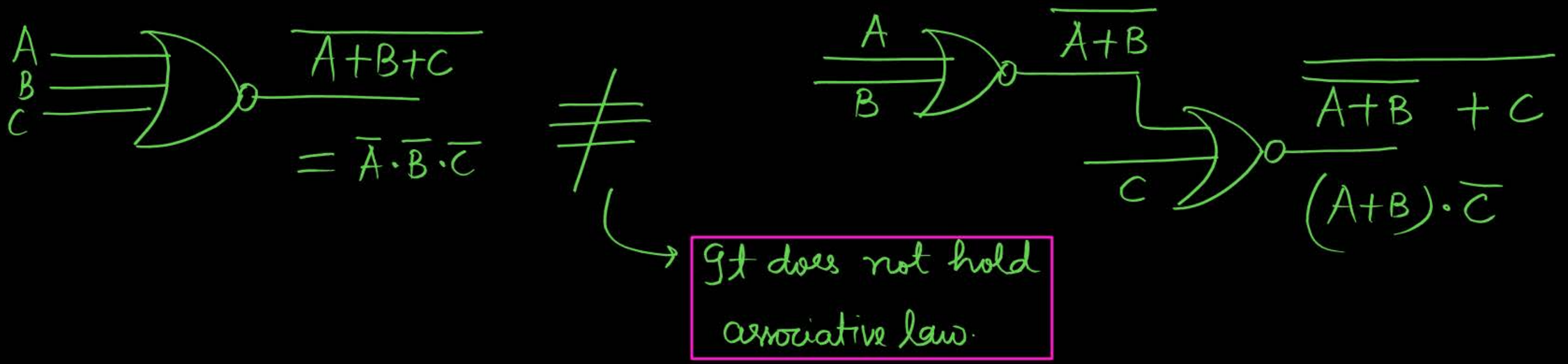
$= \overline{A}(A+\overline{B})$

$= 0 + \overline{A}\cdot\overline{B} = \overline{A}\,\overline{B}$

- Commutative Law :



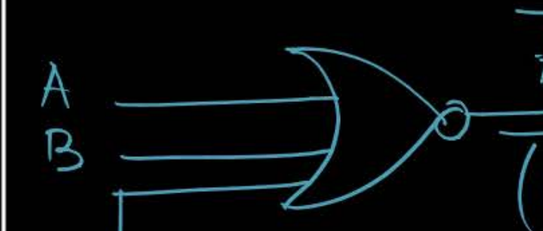
- Associative Law :



- IMP Points :

- If any one of the i/p line is at logic '1' then irrespective of other i/p lines O/P will definitely be '0'.

- O/P will be '1' only if all the i/p lines will be '0' together.

- 

$$\overline{A+B+0} = \overline{A+B}$$

$$\overline{(A+B+1)} = \overline{1} = 0$$

Control i/p = 0 [enable i/p for NOR gate]

= 1 [disable i/p for NOR gate.]

H.W.

Minimum no. 2-i/p NAND gate required to implement

1. $f(A,B) = A\bar{B} \oplus \bar{A}\bar{B}$

2. $f(A,B,D) = \overline{\bar{A}B} + \bar{B} + \bar{D}$

3. $f(A,B,C) = AB \oplus BC \oplus ABC$

4. $f(A,B,C) = \bar{A}B + A\bar{B} + BC$



2 Minute Summary

- NAND gate
- NOR gate.

Thank you

GW

Soldiers !

